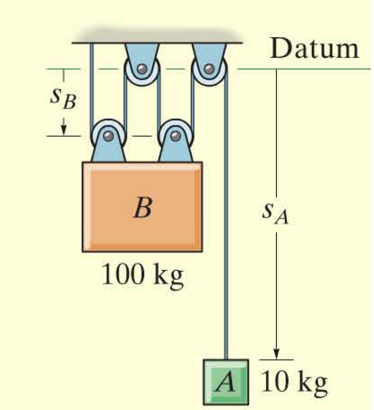
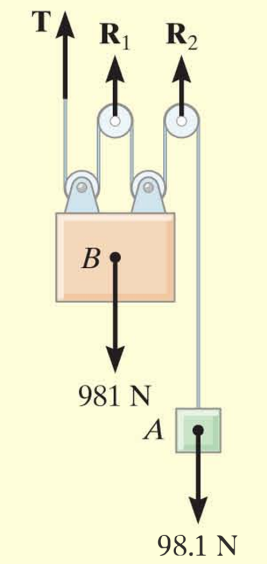
TUTORIAL 4: WORK AND ENERGY

ME 1060: DYNAMICS

Question 4

Blocks A and B shown in figure have a mass of 10 kg and 100 kg, respectively. Determine the distance B travels when it is released from rest to the point where its speed becomes 2 m/s.





Work (Free-Body Diagram). As shown on the free-body diagram of the system, Fig. 14–14*b*, the cable force **T** and reactions $\mathbf{R}_1$ and $\mathbf{R}_2$ do *no work*, since these forces represent the reactions at the supports and consequently they do not move while the blocks are displaced. The weights both do positive work if we *assume* both move downward, in the positive sense of direction of s_A and s_B.

Principle of Work and Energy. Realizing the blocks are released from rest, we have

$$\Sigma T_1 + \Sigma U_{1-2} = \Sigma T_2$$

$$\left\{\tfrac{1}{2}m_A(v_A)_1^2 + \tfrac{1}{2}m_B(v_B)_1^2\right\} + \{W_A\,\Delta s_A + W_B\,\Delta s_B\} = \left\{\tfrac{1}{2}m_A(v_A)_2^2 + \tfrac{1}{2}m_B(v_B)_2^2\right\}$$

$$\{0 + 0\} + \{98.1\text{ N}\,(\Delta s_A) + 981\text{ N}\,(\Delta s_B)\} = \left\{\tfrac{1}{2}(10\text{ kg})(v_A)_2^2 + \tfrac{1}{2}(100\text{ kg})(2\text{ m/s})^2\right\} \quad (1)$$

Kinematics. Using the methods of kinematics discussed in Sec. 12.9, it may be seen from Fig. 14–14*a* that the total length l of all the vertical segments of cable may be expressed in terms of the position coordinates s_A and s_B as

$$s_A + 4s_B = l$$

Hence, a change in position yields the displacement equation

$$\Delta s_A + 4\,\Delta s_B = 0$$

$$\Delta s_A = -4\,\Delta s_B$$

Here we see that a downward displacement of one block produces an upward displacement of the other block. Note that Δs_A and Δs_B must have the *same* sign convention in both Eqs. 1 and 2. Taking the time derivative yields

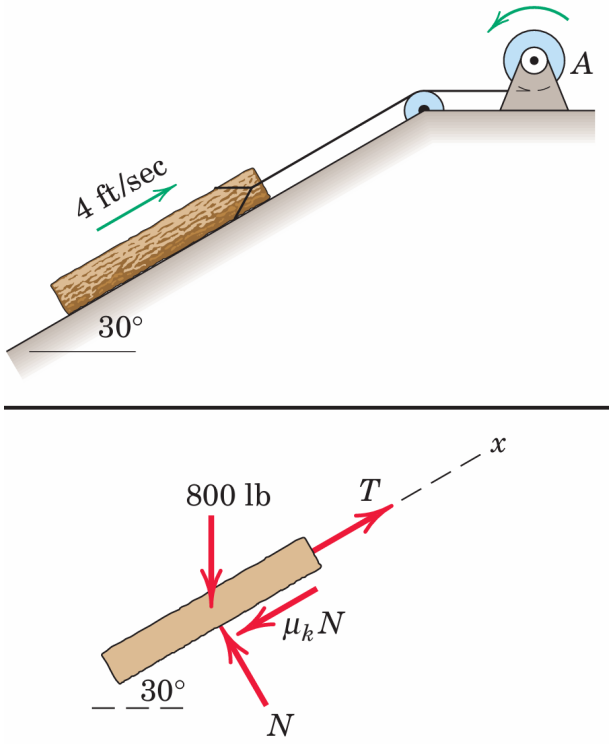
$$v_A = -4v_B = -4(2\text{ m/s}) = -8\text{ m/s} \quad (2)$$

Retaining the negative sign in Eq. 2 and substituting into Eq. 1 yields

$$\Delta s_B = 0.883\text{ m} \downarrow \qquad \textit{Ans.}$$

Question 5

The power winch A hoists the 800-lb log up the 30° incline at a constant speed of $4\ ft/sec$. If the power output of the winch is $6\ hp$, compute the coefficient of kinetic friction μ_k between the log and the incline. If the power is suddenly increased to $8\ hp$, what is the corresponding instantaneous acceleration a of the log?



Solution. From the free-body diagram of the log, we get $N = 800 \cos 30° = 693$ lb, and the kinetic friction force becomes $693\mu_k$. For constant speed, the forces are in equilibrium so that

$$[\Sigma F_x = 0] \qquad T - 693\mu_k - 800 \sin 30° = 0 \qquad T = 693\mu_k + 400$$

The power output of the winch gives the tension in the cable

$$[P = Tv] \qquad T = P/v = 6(550)/4 = 825 \text{ lb}$$

Substituting T gives

$$825 = 693\mu_k + 400 \qquad \mu_k = 0.613 \qquad \textit{Ans.}$$

When the power is increased, the tension momentarily becomes

$$[P = Tv] \qquad T = P/v = 8(550)/4 = 1100 \text{ lb}$$

and the corresponding acceleration is given by

$$[\Sigma F_x = ma_x] \qquad 1100 - 693(0.613) - 800 \sin 30° = \frac{800}{32.2} a$$

$$a = 11.07 \text{ ft/sec}^2 \qquad \textit{Ans.}$$